Dataset Documentation

Riboswitch Structures Paired with Environmental Conditions

For Environment-Aware RNA 3D Structure Prediction

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1. Overview

This document describes the curation, structure, and intended use of a dataset linking riboswitch RNA three-dimensional structures from the Protein Data Bank (PDB) with their experimental environmental conditions. The dataset is designed to support a machine learning project investigating whether environmental parameters (ion concentrations, pH, temperature, ligand presence) can predict the conformational state of RNA structures.

Current RNA 3D structure prediction models such as RhoFold+, AlphaFold3, and RoseTTAFoldNA produce static structure predictions from sequence alone, without accounting for the environmental conditions that influence RNA folding. This dataset provides the foundation for building an environment-conditioned prediction or classification model.

2. Data Sources

2.1 Primary Sources

- RCSB Protein Data Bank (rcsb.org) — 3D atomic coordinates and experimental metadata for all riboswitch structures
- Ribocentre-switch (riboswitch.ribocentre.org) — Curated database of 56 validated riboswitch classes with PDB cross-references, ligand information, and structural annotations
- Rfam Database (rfam.org) — RNA family classifications, consensus sequences, and multiple sequence alignments for riboswitch families

2.2 Secondary Sources

- Vienna RNAfold Web Server (rna.tbi.univie.ac.at) — For computing predicted secondary structures and minimum free energy values from sequences
- Published literature — For supplementary condition data not captured in PDB headers (e.g., specific Mg2+ titration experiments)

3. Data Collection Steps

Follow these steps to build the dataset. Each step includes the tools needed and expected output.

Step 1: Identify All Riboswitch PDB Entries

Go to riboswitch.ribocentre.org/structures/ and download or manually record the full list of PDB IDs for each riboswitch class. Alternatively, search RCSB PDB using the query term "riboswitch" and filter for RNA-only structures (exclude protein-RNA complexes unless the riboswitch is the primary subject).

Expected output: A spreadsheet with columns: PDB_ID, riboswitch_class, ligand_type

Tools: Web browser, Excel or Google Sheets

Step 2: Download PDB Files

For each PDB ID, download the structure file in mmCIF format from RCSB. This can be done manually from the website or programmatically:

```
import requestsfor pdb_id in pdb_ids:    url = f"https://files.rcsb.org/download/{pdb_id}.cif"    r =
requests.get(url)    with open(f"{pdb_id}.cif", "wb") as f:        f.write(r.content)
```

Expected output: A folder of .cif files, one per structure

Tools: Python with requests library

Step 3: Extract Environmental Conditions from PDB Headers

The crystallization and experimental conditions are stored in mmCIF fields. Key fields to extract:

- _exptl_crystal_grow.pdbx_details — Free-text description of crystallization buffer (contains Mg2+, ion concentrations, pH)
- _exptl_crystal_grow.pH — pH value (when reported as a separate field)
- _exptl_crystal_grow.temp — Crystallization temperature
- _diffrn.ambient_temp — Data collection temperature (usually 100K for X-ray)
- _entity.pdbx_description — Description of the RNA and any bound ligands

WARNING: The _exptl_crystal_grow.pdbx_details field is free text and highly inconsistent across PDB entries. You will need to manually parse and standardize values. Some entries will be missing data entirely. Record "NaN" or "not reported" for missing values rather than guessing.

Expected output: Updated spreadsheet with environmental condition columns added

Tools: Python with BioPython (Bio.PDB.MMCIF2Dict) or pandas for parsing

Step 4: Label Conformational State

For each structure, determine whether it represents an apo (ligand-free) or holo (ligand-bound) state. This information comes from:

- The PDB entry title and abstract (usually states explicitly)
- The presence or absence of a ligand in the coordinate file (_pdbx_nonpoly_scheme or HETATM records)
- The Ribocentre-switch annotation for that entry

- Published literature describing the structure

Expected output: A "conformational_state" column with values: apo, holo, or intermediate

Step 5: Compute Sequence and Structure Features

For each RNA sequence in the dataset:

1. Extract the RNA sequence from the PDB file or retrieve from Rfam
2. Compute sequence-level features: length, GC content, dinucleotide frequencies, and 3-mer frequencies using Python (see Feature Table below)
3. Submit the sequence to the Vienna RNAfold web server (or use the command-line tool if installed) to obtain the predicted secondary structure, minimum free energy, and structural element counts
4. Record all features in the dataset spreadsheet

Tools: Python (string operations for k-mers), Vienna RNAfold web server

Step 6: Compute Structural Similarity Metrics (Optional but Recommended)

For riboswitch classes that have both apo and holo structures, compute pairwise RMSD between all structures of the same class using US-align or PyMOL. This provides the target variables rmsd_to_apo and rmsd_to_holo for regression models.

Tools: US-align (command line) or PyMOL (GUI, good for visualization)

4. Feature Descriptions

The dataset contains three categories of features: environmental, sequence-based, and structure-based. Green rows are environmental features, yellow rows are sequence features, and blue rows are structure features.

Feature Name	Data Type	Source	Description
mg_concentration_mM	Float	PDB header	Mg2+ concentration in crystallization buffer (mM)
ph	Float	PDB header	pH of crystallization/experimental buffer
temperature_K	Float	PDB header	Temperature during experiment (Kelvin)
ligand_present	Binary (0/1)	PDB header	Whether cognate ligand was present
ligand_identity	Categorical	PDB header	Name of ligand (e.g., adenine, TPP, FMN, preQ1)
monovalent_ion_conc_mM	Float	PDB header	K+ or Na+ concentration (mM)

method	Categorical	PDB header	X-ray, NMR, or Cryo-EM
resolution_A	Float	PDB header	Structure resolution in Angstroms
sequence	String	PDB/Rfam	RNA nucleotide sequence (A, G, C, U)
seq_length	Integer	Computed	Number of nucleotides
gc_content	Float	Computed	Fraction of G and C nucleotides
dinucleotide_freq	Float array (16)	Computed	Frequency of each dinucleotide (AA, AG, AC, AU, ...)
kmer_3_freq	Float array (64)	Computed	Frequency of each 3-mer
num_base_pairs_pred	Integer	RNAfold	Predicted base pairs from Vienna RNAfold
mfe	Float	RNAfold	Minimum free energy of predicted 2D structure (kcal/mol)
num_stems	Integer	RNAfold	Number of helical stems in predicted 2D structure
num_loops	Integer	RNAfold	Number of loops (hairpin, internal, multi)
has_pseudoknot	Binary (0/1)	Literature	Whether the riboswitch class forms pseudoknots

5. Target Variables

The primary prediction target is the conformational state of the RNA. Multiple formulations are provided to support both classification and regression approaches.

Target Variable	Type	Description
conformational_state	Binary	Apo (0) or Holo (1) - primary classification target
rmsd_to_apo	Float	RMSD of structure to reference apo conformation (Angstroms)
rmsd_to_holo	Float	RMSD of structure to reference holo conformation (Angstroms)
apo_holo_score	Float (0-1)	Continuous score: 0 = fully apo, 1 = fully holo. Computed as rmsd_to_apo / (rmsd_to_apo + rmsd_to_holo)

6. Priority Riboswitch Structures

The following riboswitch classes have confirmed paired apo and holo structures in PDB and should be prioritized for dataset construction. These are ordered by priority for the project.

Riboswitch	Apo PDB	Holo PDB	RMSD	Priority
PreQ1 Class I	3Q51, 5KH8	3GCA, 6VUI	Multiple states	HIGHEST - 4 apo states
Adenine (Purine)	5E54	4TZX, 1Y26	3.6 Å	HIGH - 2 apo states
TPP	8F4O, 9C6K	2GDI	11.2 Å	HIGH - largest change
FMN	6WJR, 6WJS	3F2Q	2.1 Å	HIGH - clean pair
Glutamine	5DDO	5DDP	12.8 Å	MEDIUM - largest RMSD
THF	7KD1	4LVV	3.6 Å	MEDIUM - cooperative
SAM-I	3IQP	2GIS	Ensemble	MEDIUM
Fluoride	5KH8	3VRS	~0 Å	LOWER - same fold

Note: Additional structures with only holo conformations can be included as supplementary data points. Structures solved under different crystallization conditions (different Mg2+ concentrations, pH values, or temperatures) within the same conformational state also provide useful data points.

7. Expected Dataset Size

Based on available PDB entries for the riboswitch classes listed above:

- Core paired structures (apo + holo): approximately 20–40 data points
- Including mutant and variant structures: approximately 40–80 data points

- Including holo-only structures from additional riboswitch classes: approximately 80–150 data points
- Including structures from different crystallization conditions of the same RNA: potentially 100–200 data points

This is a small dataset by ML standards. Appropriate modeling strategies include: cross-validation (leave-one-riboswitch-class-out), simple models that require fewer data points (logistic regression, random forest, SVM), and careful regularization to prevent overfitting.

8. Output File Format

The final dataset should be saved as a CSV file with one row per PDB structure and the columns described in the Feature Table above, plus the target variable columns. A suggested filename convention is:

riboswitch_env_dataset_v1.csv

Include a companion README.txt file documenting any manual curation decisions, missing data handling, and version notes.

9. Suggested ML Workflow

1. Load the CSV dataset into a pandas DataFrame
2. Exploratory data analysis: distributions of each feature, correlation matrix, class balance (apo vs. holo counts)
3. Handle missing values: drop rows with too many NaNs; impute remaining with median or mode
4. Split into features (X) and target (y); use conformational_state for classification or apo_holo_score for regression
5. Train baseline models using scikit-learn: LogisticRegression, RandomForestClassifier, SVC, MLPClassifier
6. Evaluate using 5-fold cross-validation; report accuracy, precision, recall, F1-score, and confusion matrix
7. Analyze feature importance (from Random Forest) to determine which environmental and sequence features are most predictive
8. Compare model performance with and without environmental features to demonstrate their added predictive value

10. Key References

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